R A H U L K A R N

Enterprise Solution Architect · InsurTech & Financial Services · AI-Driven Transformation

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PROFILE

Solution architect and Digital Transformation Officer at one of Nepal's largest life insurers, owning solution architecture and application-layer security for a platform serving 90,000+ users. Backed by 9 years of hands-on .NET engineering: a premium calculation engine across 20+ product lines, payment architecture handling NPR 100B+ in transaction volume with zero duplicates, and a database program that cut query latency by ~50%. Pairs that depth with a completed MBA in Accounting & Finance and an MSc in Data Science underway at Coventry, plus professional training in software architecture, insurance, database administration, and AML/CFT.

EXPERIENCE

Solution Architect / Digital Transformation Officer

Oct 2026 – Present

Nepal Life Insurance Co. Ltd., Kathmandu · Senior Officer grade

- Core team member driving the digital transformation of Nepal Life's insurance platform (90,000+ users), with ownership of solution architecture and application-layer security.
- **Digital transformation:** digitize policy issuance, claims, and agent/customer journeys; modernize the core platform from monolith to modular, cloud-ready services.
- **Solution architecture:** design and review system architecture, data models, and API/integration contracts across policy administration, underwriting, claims, and digital channels.
- **Application security:** lead security updates and hardening of the application layer (OAuth2/OIDC, RBAC, OWASP-aligned controls, audit logging).
- **Lead development:** guide the engineering team on delivery of new modules; own code review, Git Flow, and CI/CD standards.

Senior .NET Developer / Tech Lead

Jan 2023 – Oct 2026

Nepal Life Insurance Co. Ltd., Kathmandu · Officer grade

- Lead developer for the company's core insurance platform (policy administration, underwriting, claims, and digital channels), serving 90,000+ active users.
- Shipped enterprise portal and API ecosystem with RBAC, OAuth2/OIDC authentication, and OWASP-aligned security controls.
- Drove database performance program: ~50% reduction in query latency across Oracle and SQL Server through indexing, execution-plan analysis, and stored-procedure refactoring.
- Designed payment gateway integration with idempotency keys, automated reconciliation, and audit logging; zero duplicate transactions on NPR 100B+ volume.
- Owned technical design reviews and authored RFCs covering data models, API contracts, and migration strategy for new modules.
- Established code review standards, Git Flow branching, and CI/CD pipelines; reduced release cycle from monthly to bi-weekly.
- Coordinated 6+ engineers through phased modernization (monolith to modular); mentored junior and mid-level developers on .NET Core, async patterns, EF Core performance, and clean architecture.
- Implemented observability stack (structured logging, health checks, performance counters) enabling sub-hour incident detection on production workloads.
- Researched applied AI/ML for insurance use cases: fraud detection, claims triage, and predictive underwriting.

Software Engineer

Nov 2018 – Dec 2022

Nepal Life Insurance Co. Ltd., Kathmandu · Junior Officer grade

- Built premium calculator engine across 20+ life insurance product lines; standardized actuarial rules, improving sales workflow efficiency by ~30%.
- Developed asset management application handling NPR 100B+ portfolio with audit controls and regulatory reporting.
- Built secure onboarding system for 5,000+ agents and customers with KYC workflows, eligibility validation, and payment gateway integration.
- Built API integrations connecting underwriting, policy issuance, and claims across core insurance systems.
- Developed scheduled batch jobs for premium notices, policy renewals, and commission calculations, processing 100K+ records per cycle.
- Migrated legacy ASP.NET WebForms modules to ASP.NET MVC / Web API, improving maintainability and front-end responsiveness.

Software Engineer

Oct 2017 – Nov 2018

Innovative Ghar Nepal, Kathmandu

- Built multi-tenant real estate platform (ASP.NET MVC, C#) with listings, tenant management, RBAC, and reporting modules.
- Designed normalized SQL Server schema and stored procedures backing search, filtering, and tenant isolation.

OPEN SOURCE

grahan 2026

zero-dependency TypeScript sky engine · github.com/svarbhanu/grahan

- Three MIT-licensed npm packages for astronomy, panchang, and Bikram Sambat conversion; v1.0 with a frozen public API and 279 tests from Singapore to the Arctic.
- Verified against Swiss Ephemeris: Sun longitude within 4.6″, sunrise within 4.6 s, and all 909 eclipses (1900–2100) detected and classified exactly.
- Released from CI through npm trusted publishing with signed provenance; no long-lived publish credentials.

TECHNICAL SKILLS

Architecture & APIs: Solution architecture, Clean Architecture, domain-driven design, modular architecture, service decomposition, event-driven workflows, REST API design, API versioning, OAuth2/OIDC

Languages & Frameworks: C#, .NET 10 / .NET 8 LTS, ASP.NET Core, Entity Framework Core, Web API, MediatR; TypeScript; Python (pandas, scikit-learn)

Databases: Oracle, SQL Server, T-SQL; SQL Server administration, query optimization, execution plans, indexing strategies

Data & AI/ML: Python data stack, ML.NET, Semantic Kernel; applied to insurance use cases

Cloud & DevOps: Azure (App Services, SQL, Key Vault), Docker, Terraform, Git, CI/CD pipelines, OpenTelemetry, deployment automation

Leadership & Delivery: Digital transformation, Agile project management, design reviews and RFCs, mentoring, code review standards

Insurance Domain: Policy onboarding, product information, underwriting, claims, premium calculation, accounting, KYC, AML/CFT, payment reconciliation, agent/distribution

EDUCATION

MSc, Data Science & Computational Intelligence, Coventry University, UK (Distance, Softwarica) Expected Feb 2027

MBA, Accounting & Finance, Indira Gandhi National Open University Completed Sept 2026

BE, Computer Science & Engineering, Advanced College of Engineering and Management, Tribhuvan University 2013 – 2017

PROFESSIONAL TRAINING

- **Mastering in Software Architecture** · Kathmandu University, 2026; advanced patterns, enterprise systems, and cloud-native architecture
- **Insurance Fundamentals** · Insurance Institute of Nepal; life insurance principles, products, and practice
- **Microsoft SQL Database Administrator** · SQL Helps, 2021; SQL Server administration, backup and recovery, performance tuning
- **AML/CFT** · FIU-Nepal / Nepal Rastra Bank; anti-money-laundering and counter-financing-of-terrorism compliance

PUBLICATIONS

IEEE Publication: “Attendance and Security Assurance using Image Processing,” ICCMC 2018, IEEE Xplore